

Yuvraj Singh Naruka

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EDUCATION

Vellore Institute of Technology, Bhopal

Bachelor of Technology in Computer Science and Engineering , CGPA: 7.51/10

Bhopal, Madhya Pradesh

Oct 2022 – Present

Kendriya Vidyalaya No.2 Jaipur Cantt.

Class XII (CBSE) , Percentage: 89.0%

Jaipur, Rajasthan

Apr 2020 – May 2021

Kendriya Vidyalaya No.2 Jammu Cantt.

Class X (CBSE) , Percentage: 90.2%

Jammu, Jammu and Kashmir

Apr 2018 – May 2019

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL

Web Development: HTML, CSS, FastAPI, REST APIs

Frameworks and Libraries: SQLAlchemy, Pydantic, Jinja2, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib

TECHNICAL PROJECTS

Blog Management System | *Python, FastAPI, SQLAlchemy, SQLite*

2026

- Developed a RESTful Blog Application using FastAPI, implementing JWT authentication, OAuth2, and 10+ CRUD API endpoints.
- Engineered asynchronous database interactions using SQLAlchemy ORM with SQLite, Pydantic validation, One-to-Many relational mapping, and asynchronous request handling.
- Integrated Jinja2 server-side templating, static file serving, custom HTML/JSON exception handling, and secure password hashing for user authentication.

Disaster Early Warning and Relief Resource Management System | *Python, Keras, TensorFlow*

2026

- Directed the development of a Custom Convolutional Neural Network (CNN) to classify disaster scene images across multiple categories, achieving automated situational awareness for emergency response teams.
- Engineered and augmented a disaster image dataset through resizing, normalization, and transformations — including rotation (180°), zoom, width/height shifts, and horizontal/vertical flips — to strengthen model generalization.
- Optimized training stability with Batch Normalization, Dropout, and adaptive learning rate scheduling via ReduceLROnPlateau and EarlyStopping, using stratified train/validation/test splits for reliable evaluation.

Pneumonia Detection | *Python, Keras, TensorFlow*

2025

- Led a team of 5 members to develop a Convolutional Neural Network (CNN) using Python, Keras, and TensorFlow to classify pneumonia from chest X-ray images, achieving 91.1% accuracy on test set.
- Processed and augmented a dataset of 5,000+ Chest X-ray images, performing resizing, normalization, and image transformations to improve model generalization.
- Boosted model performance by integrating Batch Normalization, Dropout, and learning rate scheduling, yielding high sensitivity and specificity in the final deployed model.

Loan Creditworthiness Prediction | *Python, Scikit-learn*

2024

- Developed machine learning model (Random Forest Algorithm) using Python and Scikit-learn to predict loan creditworthiness, achieving optimal performance through RandomizedSearchCV with 5-fold cross-validation.
- Transformed loan application data across 18 features by implementing preprocessing pipelines with standardization and one-hot encoding, enhancing model reliability and input consistency.
- Refined credit risk predictions using class weighting (10:1 ratio) and probability scoring, optimizing for F1-score to improve classification performance on imbalanced datasets.

CERTIFICATIONS

Oracle Cloud Infrastructure Foundations Associate - *Oracle University*

Oracle Cloud Infrastructure AI Foundations Associate - *Oracle University*

EXTRACURRICULAR ACTIVITIES

Corresponding Author — *Infectious Diseases Now Journal*

2025

- Authored and published editorial "Balancing AI accuracy and medical guidance: Evaluating ChatGPT's role in UTI-related queries" in a peer-reviewed medical journal (Volume 55, Issue 3).
- Contributed to academic discourse on AI applications in healthcare and infectious disease research, addressing ethical implications and limitations of AI-driven medical consultations.

AIEM Club — *Member*

2024

KVS Jammu Regional Basketball Team — *Vice Captain*

2019